

Covariance Matrix for a Sample

Given n samples of p variables,
the sample covariance matrix is

$$\hat{\Sigma} = \frac{1}{n-1} (X - \bar{X})^T (X - \bar{X})$$

or $\text{centered } X$

$$\hat{\sigma}_{ij} = \sum_{k=1}^n (x_{ki} - \bar{x}_i)(x_{kj} - \bar{x}_j)$$

eg: P_1 P_2

$$X = \begin{bmatrix} 1 & 4 \\ 3 & 6 \end{bmatrix}$$

$$\bar{x}_1 = \frac{1+3}{2} = 2$$

$$\bar{x}_2 = \frac{4+6}{2} = 5$$

$$\bar{X} = \begin{pmatrix} \bar{x}_1 \\ \bar{x}_2 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix}$$

$$X - \bar{X} = \begin{bmatrix} 1-2 & 4-5 \\ 3-2 & 6-5 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix} \quad (X \text{ is centered})$$

$$\hat{\Sigma} = \frac{1}{2-1} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$\text{Var}(P_1) \rightarrow \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \quad \downarrow \quad \begin{bmatrix} (X - \bar{X})^T & (X - \bar{X}) \end{bmatrix}$$

$$\text{Cor}(P_1, P_2) \quad \text{Var}(P_2)$$